

MADGCN: A Meteorology-Aware Spatio-Temporal Graph Convolution Network for Long-Term Air Pollution Forecasting

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Abstract—Air quality forecasting has attracted increasing attention as global air pollution worsens. Spatiotemporal graph neural networks have become a leading paradigm, thanks to their ability to capture complex spatial and temporal dynamics in Air Quality Index (AQI) data. However, existing methods remain limited by weak modeling of long-range temporal dependencies and insufficient integration of meteorological factors. Building on a publicly available nationwide air quality dataset spanning eight years, we propose MADGCN, a Meteorology-Aware Decoupled Spatio-Temporal Convolutional Network that jointly addresses long-horizon temporal modeling and meteorological context fusion. MADGCN includes a dynamic causality discovery module grounded in Granger causality, which captures time-varying causal relationships between meteorological conditions and AQI dynamics. The inferred causal structures further guide a causal graph convolution module and a PatchMixer module, enabling effective spatial interaction modeling and multiscale temporal dependency learning. Extensive experiments against 16 strong baselines show that MADGCN achieves competitive performance for long-horizon air pollution forecasting and generalizes well under high-pollution regimes..

Index Terms—Spatiotemporal data mining, air pollution prediction, spatiotemporal graph convolutional network.

I. INTRODUCTION

AIR pollution has emerged as a critical environmental challenge of global concern. Among the various pollutants, fine particulate matter (e.g., PM_{2.5}) poses particularly severe risks to human health, ecological stability, and socioeconomic

systems [1]. Consequently, accurate air pollution forecasting—defined as the task of predicting future Air Quality Index (AQI) based on historical data—has attracted growing interest from both academia and industry. Reliable predictions can empower governments, organizations, and individuals to implement timely interventions, thereby mitigating health risks, improving environmental governance, and supporting sustainable urban development [2], [3].

Early approaches for air pollution forecasting predominantly relied on classical statistical learning methods, such as Autoregressive Integrated Moving Average (ARIMA). While these models offer simplicity and interpretability, they often fail to adequately capture the intricate nonlinear spatiotemporal patterns within AQI data, thereby limiting their accuracy. In recent years, the rapid development of deep learning has catalyzed a paradigm shift toward neural network-based forecasting methods. Among these, Spatiotemporal Graph Neural Networks (STGNNs) have emerged as particularly effective frameworks for modeling complex environmental dynamics [4], [5], [6], [7]. In STGNNs, the air quality monitoring infrastructure is represented as a graph, with stations modeled as nodes and spatial dependencies encoded through edges that evolve over time. These architectures integrate graph convolutional networks to model spatial correlations among sensor locations, while incorporating temporal modules—such as LSTM or TCN—to capture dynamic temporal transitions across multiple time horizons. Despite achieving state-of-the-art performance in various air pollution forecasting benchmarks, existing STGNN-based approaches still face several critical limitations that warrant further investigation.

1) Inefficient Modeling for Long-Term Series with Concept Drift. Most existing STGNNs operate at a single temporal scale, which limits their ability to capture the complex, long-range dependencies inherent in long-term sequences. Consequently, their forecasting performance often deteriorates substantially as the prediction horizon increases. As illustrated in Fig. 1(a), the AQI exhibits multiple temporal components, particularly seasonality and trend. Seasonality captures recurring periodic patterns that are critical for predicting air-quality dynamics. In contrast, the trend component reflects gradual shifts in the underlying AQI distribution, inducing concept drift—i.e., changes in the statistical properties of the series over time. Such drift is further amplified by evolving environmental regulations

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Code is available at our official repository

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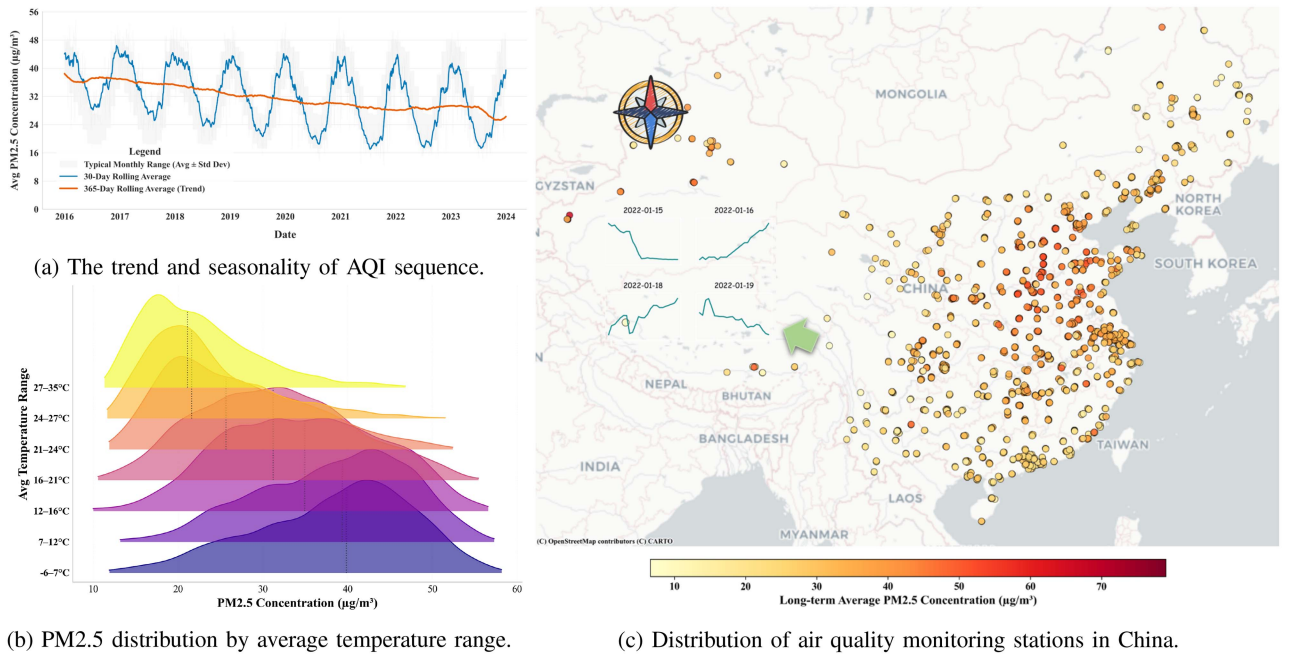


Fig. 1. Explanation of the limitations of existing air pollution forecasting models.

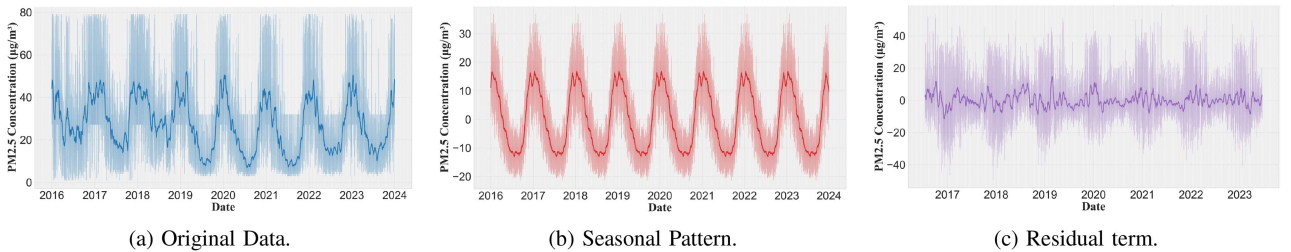


Fig. 2. Visualization of the decomposition of the AQI series.

and changing climate patterns (we visualize this phenomenon in Fig. 4). This non-stationarity poses a major challenge to generalization and accuracy for models that rely on single-scale temporal representations.

② **Underestimated meteorological factors.** Meteorological factors exert significant and causal influences on air quality. For example, as shown in Fig. 1(b), lower temperatures are associated with elevated PM2.5 concentrations—this correlation is largely attributed to increased fossil fuel combustion for heating during cold periods, which intensifies air pollution. Wind direction also plays a critical role by transporting pollutants from one region to another, thereby increasing pollution levels in down-wind areas. These intra-station and inter-station meteorological effects reflect strong causal dependencies, which are particularly beneficial for modeling high-pollution episodes. However, most existing models underutilize these causal relationships. The majority of current approaches treat meteorological factors as optional auxiliary features, limiting their ability to effectively capture the underlying spatiotemporal dynamics of air pollution propagation.

TABLE I
NOTATIONS AND EXPLANATIONS FOR CORE FACTORS

Notations	Explanations
$\mathbf{X}$	Observed AQI data
$\mathbf{Z}$	Observed meteorological data
$\mathbf{T}$	PM2.5 Trend component
$\mathbf{S}_{enh}$	PM2.5 Enhanced Seasonal component
$\mathbf{R}$	PM2.5 Residual component
$\mathbf{C}(t)$	Dynamic causal strength matrix at time step t
$\mathbf{A}_p$	Adjacency matrix of Physical Proximity Graph (PPG)
$\mathbf{A}_c^t$	Adjacency matrix of Causal Aware Graph (CAG) at time step t
$\tilde{\mathbf{X}}_s$	Fused spatial features tensor
$\hat{\mathbf{Y}}$	Predicted PM2.5 values
L_h/L_f	Lookback/prediction window lengths
N	The number of nodes

③ **Lack of High-quality Benchmark.** Existing open-source air quality datasets (e.g., KnowAir and KDDCUP-Beijing) are limited in both temporal span and spatial scale: KnowAir ends in 2017 and KDDCUP-Beijing in 2018, and each typically includes

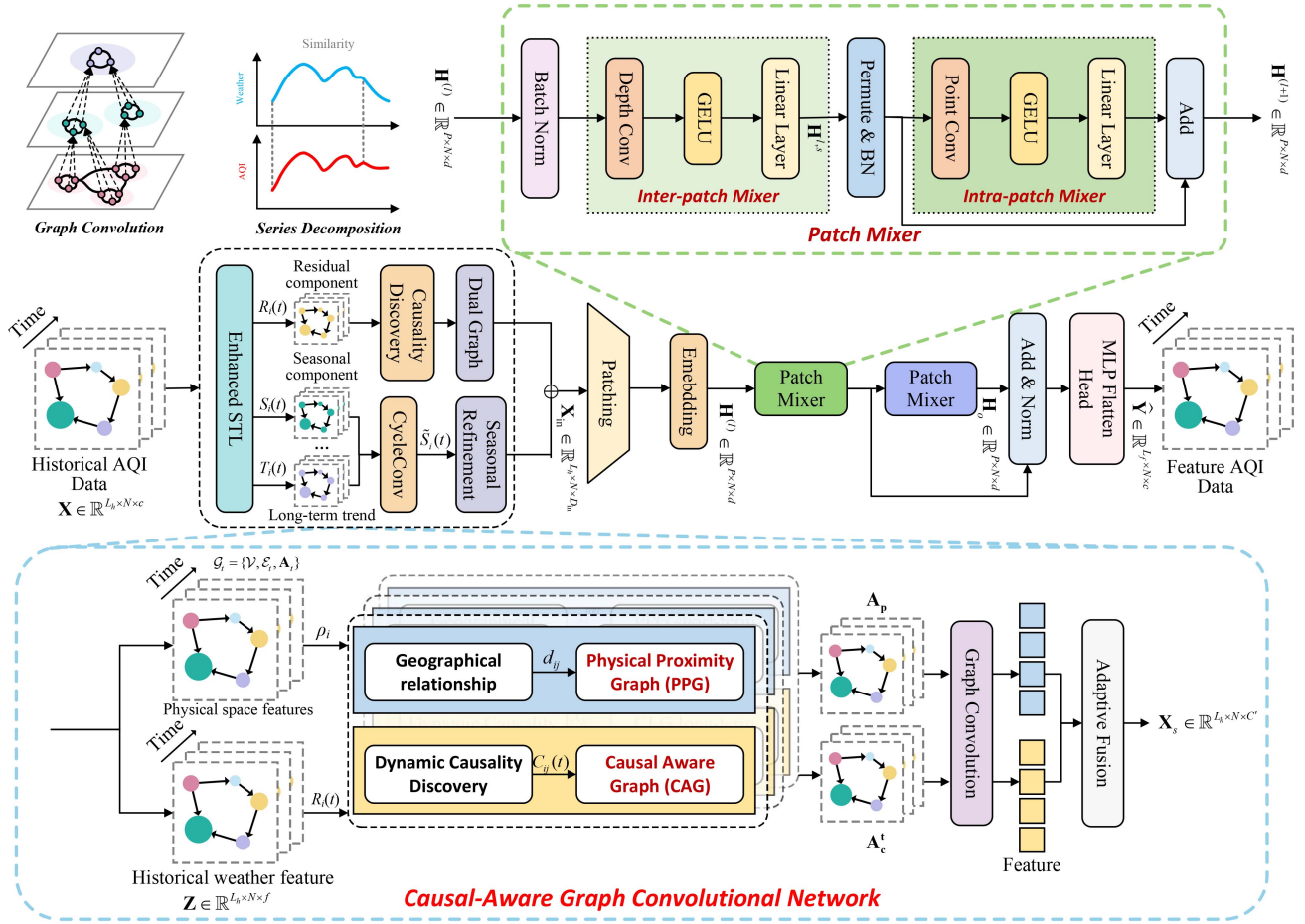


Fig. 3. The overall architecture of our MADGCN.

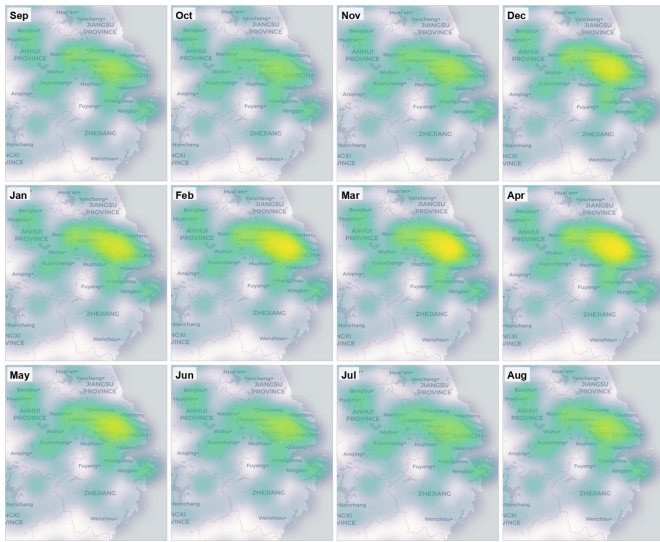


Fig. 4. Monthly PM2.5 concentration heatmaps for the Yangtze River Delta region (Sep. 2022-Aug. 2023), derived from the LargeAQ dataset.

fewer than 200 monitoring stations, as shown in Table II. Consequently, models developed on these small-scale, short-horizon benchmarks may report overly optimistic results yet generalize

TABLE II
SUMMARY OF AIR QUALITY DATASETS. OPEN ACCESS INDICATES PUBLIC AVAILABILITY, WITH THE × SYMBOL DENOTING NON-OPEN DATASETS.

Dataset/Work	#Stations	Time Span	Timesteps	Granularity	Coverage	Open Access
Beijing [1]	12	12/05/2014-31/12/2017	5856	1 h	City	✓
ShangHai [1]	8	12/05/2014-31/12/2017	5856	1 h	City	✓
ChongQing [1]	22	12/05/2014-31/12/2017	5856	1 h	City	✓
KnowAir [46]	184	01/01/2014-31/12/2018	11688	3 h	Region	✓
AirFormer [1]	1085	01/01/2015-31/12/2018	—	1 h	Nation	×
SSH-GNN-BTH [3]	479	01/01/2018-01/04/2020	—	1 h	Region	×
LargeAQ (Ours)	1341	01/01/2015-31/12/2023	70128	1 h	Nation	✓

poorly to large-scale, long-horizon forecasting, risking failures in real-world deployment. This motivates an urgent need for a national-scale benchmark with up-to-date data and standardized evaluation protocols.

To overcome the aforementioned limitations, we propose a **M**eteorology-**A**ware **D**ecoupled **G**raph **C**onvolution **N**etwork (MADGCN), a novel deep learning framework for air pollution forecasting. MADGCN integrates time series decomposition techniques with meteorology-informed causal discovery mechanisms to enable more accurate, robust, and interpretable long-term air pollution forecasting.

For temporal decomposition, we employ the Seasonal Decomposition of Time Series by Loess method. This robust non-parametric approach iteratively decomposes the time series into three interpretable components: trend, seasonality, and residual noise. Through this fine-grained decomposition, the model is better able to capture precise distributional evolution patterns, thereby effectively addressing concept drift in long-term air pollution forecasting. We further refine the representation of the seasonal component through additional modeling strategies to better capture seasonal dependencies. To model both intra-node and inter-node causal relationships, we introduce a dynamic causal discovery mechanism grounded in Granger causality. This approach infers directed interactions among spatial nodes by leveraging historical meteorological factors and AQI observations. Specifically, it formulates the causal relationship as a directed graph, where edges represent statistically inferred causal influences between locations. Causal strength is estimated via least squares regression. Building upon this causal graph, we design a novel causal-aware graph convolution module to effectively model spatiotemporal dependencies. This module incorporates two complementary graph structures: (1) the Physical Proximity Graph, which encodes geospatial adjacency, and (2) the Causal-Aware Graph, derived from the discovered causal relationships. By jointly considering these two perspectives, the model captures both physical diffusion mechanisms and environmental causal pathways. Accurate modeling of meteorological factors also helps models perceive concept drift. Finally, we introduce a Patch-Mixing module to model the temporal evolution of AQI sequences. This module operates on segmented time series patches, learning both local temporal patterns and global temporal dependencies through cross-patch interactions. It enhances the model's capacity to capture complex temporal dynamics at multiple scales, improving overall forecasting accuracy and robustness.

Our contributions can be summarized as follows:

- ① *New Insight:* We incorporate temporal structure decoupling and meteorological factors to empower long-term air pollution forecasting, leading to the design of a Meteorology-Aware Decoupled Spatio-Temporal Graph Convolutional Network (MADGCN).
- ② *Practical Solution:* For temporal decomposition, we propose a novel temporal decoupling mechanism, together with a dynamic causal-aware strategy based on the Granger causality Principle, enabling the model to learn latent causal dependencies between meteorological factors and AQI.
- ③ *Empirical Study:* Comprehensive experiments on real-world datasets against 16 advanced baselines demonstrate the effectiveness of our method in achieving superior accuracy for long-term air pollution forecasting.

We review the recent advances in air pollution forecasting in Section II, and provide detailed explanations of our framework in Section IV. Subsequently, in Section V, we present extensive experimental results to validate the effectiveness of our model. Finally, we conclude the paper and discuss potential directions for future work in Section VI and Section VII.

II. RELATED WORK

A. Time Series Prediction

PM2.5 concentration forecasting is a fundamental problem in time series prediction [8], [9], [10], [11], [12], [13]. Early studies primarily relied on statistical methods such as ARIMA [14], which are capable of modeling linear temporal dependencies but struggle to capture the nonlinear dynamics that are typical of air pollution processes. Deep learning approaches, particularly long short-term memory (LSTM) networks [15], have significantly improved the modeling performance of PM2.5 time series [16]. Meanwhile, to address the challenges of long-horizon time series forecasting, a variety of innovative methods have been developed [17], [18], [19]. In particular, Transformer-based models [20], [21] have gradually become mainstream, with variants such as Informer [22], Autoformer [23], and FEDformer [24] introducing sparse attention mechanisms and frequency domain analysis to efficiently handle long sequences. PDF [8] further advances long-term forecasting by explicitly decoupling periodic and non-periodic components. SparseTSF [25] enhances parameter efficiency by decoupling trend and seasonal components. Despite these advances, purely temporal models often overlook the crucial spatial correlations inherent in the diffusion and transport processes of air pollutants.

B. Spatiotemporal Graph Convolutional Network

Air quality at a specific location is often influenced by surrounding regions, making spatiotemporal modeling crucial for accurate forecasting [26]. Early deep learning approaches employed convolutional neural networks (CNNs) for spatial feature extraction and combined them with long short-term memory (LSTM) networks to model temporal dynamics. More recently, spatiotemporal graph neural networks (STGNNs) have become the dominant choice for capturing spatiotemporal correlations [27], [28], [29], [30], [31]. Early GNN-based methods typically construct static spatial graphs based on geographical proximity or fixed distance thresholds. However, such representations often fail to capture the complex transport and diffusion patterns of pollutants. To address these limitations, dynamic graph approaches allow spatial dependencies to evolve over time [32], [33]. Some models leverage attention mechanisms to learn data-driven adaptive adjacency matrices [34], while others focus on modeling multi-step dependencies to further improve forecasting accuracy [35].

C. Dedicated Air Pollution Forecasting Model

Early air pollution forecasting models rely on physics-based simulations and domain-specific knowledge. With the expansion of urban-scale sensor networks and advances in machine learning, data-driven models have emerged as a promising alternative. Recent progress in machine learning—particularly deep learning—has significantly improved forecasting accuracy [1], [36]. Inspired by the success of spatiotemporal graph convolutional networks, researchers have developed specialized frameworks for air pollution forecasting based on GCNs [37], [38]. These models typically represent air quality data as a

graph, where temporal dependencies are captured using RNNs or Transformers, and spatial interactions are modeled through GCN or attention mechanisms. For instance, AirFormer [1] improves upon the Transformer architecture by introducing a two-stage framework that efficiently captures spatiotemporal dynamics. Another line of research focuses on physics-informed modeling via partial differential equations (PDEs), aiming to uncover the underlying dynamics of AQI evolution [39], [40], [41]. For example, Air-DualODE [40] incorporates a discretized boundary-aware advection-diffusion equation to simulate pollutant transport. However, solving PDE-based models incurs high computational costs, which limits their scalability in large-scale monitoring systems.

III. PRELIMINARY STUDY

Spatiotemporal Dynamic Graph: We represent the air quality monitoring system at time-step t as a graph $\mathcal{G}_t = \{\mathcal{V}, \mathcal{E}_t, \mathbf{A}_t\}$ where $\mathcal{V}$ is the node set with N nodes (e.g., monitoring stations) and $\mathcal{E}_t$ is the edge set. $\mathbf{A}_t$ is a time-varying adjacency matrix representing the dynamic relationships between nodes at time-step t , and we will explain its construction details below.

AQI and Meteorology: We use $\mathbf{x}_t \in \mathbb{R}^{N \times c}$ to denote the air quality index collected from N stations at time-step t , where c is the number of features such as PM_{2.5} or NO₂. Meteorological features $\mathbf{z}_t \in \mathbb{R}^{N \times f}$ are collected from all stations where f is the number of meteorological features such as temperature and wind speed.

Problem Definition: Given observed AQI $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_{L_h}\} \in \mathbb{R}^{L_h \times N \times c}$ and meteorological features $\mathbf{Z} = \{\mathbf{z}_1, \dots, \mathbf{z}_{L_h}\} \in \mathbb{R}^{L_h \times N \times f}$ over the past L_h time steps, air pollution forecasting aims to predict the AQI for the next L_f time steps $\mathbf{Y} = \{\mathbf{x}_{L_h+1}, \dots, \mathbf{x}_{L_h+L_f}\} \in \mathbb{R}^{L_f \times N \times c}$.

IV. METHOD

In this section, we describe the details of MADGCN, as shown in Fig. 3 and Algorithm 1. Some important variables and their definitions are provided in Table I.

A. Time Series Decomposition

Time series decomposition techniques are currently popular in the time series community, as they allow models to independently model different components of the time series, enabling more accurate insights. Here, we adopt the Seasonal Decomposition of Time Series by Loess [42] for time series decomposition. Specifically, the input AQI time series of node v_i at time step t , denoted as $X_i(t) \in \mathbb{R}^c$ (extracted from $\mathbf{X}$), can be decomposed into the following components:

$$X_i(t) = T_i(t) + S_i(t) + R_i(t), \quad (1)$$

where T_i denotes the trend component of the node v_i , reflecting the long-term evolution of AQI; S_i represents the seasonal component, capturing recurring and periodic patterns; R_i corresponds to the residual component, modeling short-term fluctuations and irregularities. This decomposition is tailored to

mitigate concept drift: the trend component $T(t)$ explicitly models long-term distributional shifts, allowing the model to adapt to systematic changes in air-quality dynamics over multi-year horizons. A decomposition case is visualized in Fig. 2. The STL decomposition method performs r iterations to accurately separate the individual components (trend, seasonal, and residual) from the original time series. Specifically,

Step 1. Initialization: We perform the Robust Initial Trend Estimation to initialize the trend component, $T_i^{(0)}(t)$ using a robust local regression (LOESS) approach:

$$T_i^{(0)}(t) = \arg \min_T \sum_{k=1}^{L_h} \rho(X_i(k) - T(k)) \quad (2)$$

where $\rho(\cdot)$ is a robust loss function designed to minimize the influence of outliers, and the summation is over all time points of the series. We also initialize the seasonal component, for instance, $S_i^{(0)}(t) = 0$.

Step 2. Iterative Optimization: The subsequent steps form an iterative refinement process for the r -th iteration step. This step begins by detrending the series using the current trend estimate $T_i^{(r)}(t)$. The detrended series $Z_i^{(r)}(t)$ is:

$$Z_i^{(r)}(t) = X_i(t) - T_i^{(r)}(t) \quad (3)$$

The dominant seasonal period, p^* , is then identified from this detrended series $Z_i^{(r)}(t)$ by finding the lag that maximizes the autocorrelation:

$$p^* = \arg \max_p \text{AutoCorr}(Z_i^{(r)}(t), p) \quad (4)$$

where the maximization is over a set of relevant candidate periods. Following the identification of p^* , the seasonal component for the next iteration, $S_i^{(r+1)}(t)$, is extracted by applying LOESS smoothing to the seasonal sub-series of $Z_i^{(r)}(t)$.

Using the updated seasonal component $S_i^{(r+1)}(t)$, the series is first deseasonalized:

$$D_i^{(r+1)}(t) = X_i(t) - S_i^{(r+1)}(t) \quad (5)$$

Then, the trend component for the next iteration, $T_i^{(r+1)}(t)$, is re-estimated by applying LOESS smoothing to this deseasonalized series $D_i^{(r+1)}(t)$:

$$T_i^{(r+1)}(t) = \arg \min_T \sum_{k=1}^{L_h} \rho(D_i^{(r+1)}(t) - T(t)) \quad (6)$$

This iterative process is repeated for a predetermined number of iterations, or until the components $T_i(t)$ and $S_i(t)$ converge. Let $T_i(t)$ and $S_i(t)$ denote the final estimates after the iterations are complete.

Step 3. Extract Residual Term: The residual term $R_i(t)$ is calculated based on the final estimates of the trend $T_i(t)$ and seasonal $S_i(t)$ components:

$$R_i(t) = X_i(t) - T_i(t) - S_i(t) \quad (7)$$

Step 4. Seasonal Component Enhancement: The initially extracted seasonal component $S_i(t)$ is further refined using a dedicated CyclConv module from prior works [43], [44] and

a specialized Mixer module, aiming to better capture and reconstruct the underlying seasonal patterns. The computation proceeds as follows:

$$\tilde{S}_i(t) = S_i(t) + \gamma \left(W_c \cdot \text{CyclConv}(S_i(t), \Theta_{\text{cycle}}) + b_c \right) \quad (8)$$

where Θ_{cycle} represents the parameters of the CyclConv module. W_c and b_c are learnable parameters, and γ is a tuning hyperparameter.

The final outputs of this decomposition stage are the trend component $T_i(t)$, the enhanced seasonal component $S_{\text{enh},i}(t)$, and the residual component $R_i(t)$.

B. Dynamic Causality Discovery With Granger Causality

Meteorological factors exert causal influences on the AQI, which can be categorized into two types: inter-node causal relationships and intra-node causal effects. For instance, wind can transport pollutants from one region to another, thereby affecting local air quality—this is defined as an inter-node causal relationship. On days with lower temperatures, pollutant levels tend to rise across multiple locations, primarily due to increased fossil fuel combustion for heating, leading to elevated AQI values (see Fig. 1(b)). This type of influence is referred to as an intra-node causal effect, as it reflects a common meteorological impact that operates independently at each node. To model these causal interactions, we propose a dynamic causality discovery method grounded in Granger causality, which utilizes a directed causal graph to capture how meteorological conditions influence air quality over time.

1) *Granger Causality Test*: The Granger causality Test is a statistical method widely used in time series analysis [45]. Its core idea is to determine whether one variable can predict the future evolution of another by examining their temporal lag relationships. Under Granger's formalism, a variable $\mathbf{X}$ is said to Granger-cause another variable $\mathbf{Y}$ if the inclusion of $\mathbf{X}$'s past values significantly improves the prediction of $\mathbf{Y}$'s future values beyond what is achievable using only $\mathbf{Y}$'s own history. Specifically, for two time series $\mathbf{X}$ and $\mathbf{Y}$, the Granger causality Test relies on an autoregressive model:

$$\mathbf{Y}_t = \beta + \sum_{k=1}^{K_1} b_k \mathbf{Y}_{t-k} + \sum_{k=1}^{K_2} c_k \mathbf{X}_{t-k} + \varepsilon_t^2 \quad (9)$$

where K_1 and K_2 are lag orders. ε_t^1 and ε_t^2 are zero-mean white noise. In the second model, lagged terms of $\mathbf{X}_{t-k}$ are introduced as additional explanatory variables. If the inclusion of these lagged terms leads to a statistically significant improvement in prediction accuracy, we conclude that $\mathbf{X}$ influences $\mathbf{Y}$. To this end, we discover the inter-node and intra-node causal relationships between meteorological factors and AQI, which are represented by a graph adjacency matrix $C(t) \in \mathbb{R}^{N \times N}$, where $C_{i,j}^k(t)$ models how the k -th meteorological factor, originating from the node v_i , affects node v_j at time step t .

Following this idea, we employ an autoregressive approach to compute $C_{i,j}^k(t)$. Specifically, classical Granger causality assumes linear autoregression; thus we apply it only to the residual term, which is more stationary after removing trend/seasonality;

the remaining nonlinear spatiotemporal patterns are captured by the subsequent neural modules. We select the residual term of node v_j and the k -th meteorological variable of node v_i , which are denoted as $R^j \in \mathbb{R}^{T \times c}$ and $Z^{i,k} \in \mathbb{R}^{T \times f}$. We incorporate these two variables into the Granger causality framework, in accordance with the Granger Causality Principle as follows,

$$R_t^j = v + \sum_{p=1}^{P_R} \alpha_p R_{t-p}^j + \sum_{q=1}^{P_K} \beta_q Z_{t-q}^{i,k} + \epsilon_t \quad (10)$$

where R_t^j denotes the residual term of node v_j at time step t . v is the intercept term, and α_p and β_q are coefficients. P_R and P_K denote the lagged terms of R^j and $Z^{i,k}$, respectively. ϵ_t is the error term. We use matrix form to express the above formula as follows,

$$\mathbf{Y}^j = \mathbf{X}\Theta + \mathbf{E} \quad (11)$$

where we use the previous $(N_t - 1)$ time steps of each time step as additional features, i.e., $\mathbf{Y}^j \in \mathbb{R}^{N_t} = [R_{t-N_t}^j, \dots, R_{t-1}^j, R_t^j]$. Similarly, $\mathbf{X} \in \mathbb{R}^{N_t \times (P_R + P_K)}$ includes the intercept term, the residual component R_t^j , the lagged terms of meteorological factors $Z_t^{i,k}$, as well as their neighboring N_t time steps, and $\mathbf{E} \in \mathbb{R}^{N_t \times 1}$ represents the corresponding error term. $\Theta = [\beta_0, \alpha_1, \dots, \alpha_{P_R}, \beta_1, \dots, \beta_{P_K}]$ is the coefficient matrix. The estimated coefficient vector $\hat{\Theta}$ is obtained via Ordinary Least Squares (OLS), and is given by the following expression:

$$\hat{\Theta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}^j \quad (12)$$

From the estimated vector $\hat{\Theta}$, we can extract the coefficient estimates associated with lagged terms of meteorological factors $\hat{\beta} = [\hat{\beta}_1, \dots, \hat{\beta}_{P_K}]$. Finally, $C_{i,j}^k(t)$ can be computed as the weighted arithmetic mean of the absolute values of the estimated regression coefficients for its lagged terms:

$$C_{i,j}^k = \sum_{q=1}^{P_K} w_q |\hat{\beta}_q| \quad (13)$$

where w_q is a non-negative weight assigned to the absolute value of the coefficient for the q -th lagged term with $\sum_{q=1}^{P_K} w_q = 1$. Finally, we aggregate the f meteorological factors to obtain the final causal strength from node v_i to node v_j at time step t :

$$C_{ij}(t) = g(\{C_{ij}^k(t)\}_{k=1}^f), \quad (14)$$

where $g(\cdot)$ represents a deterministic aggregation function, a weighted average reflecting the relative importance of each factor. The resulting $C_{ij}(t)$ quantifies the overall causal influence directed from node v_i to node v_j , as driven by the considered meteorological factors.

C. Causal-Aware Graph Convolutional Network

To effectively capture the complex spatial dynamics inherent in AQI, we construct and use two different graph structures.

1) *Physical Proximity Graph*: Stations that are geographically close often exhibit similar AQI patterns. To capture this spatial dependency, we construct a physical proximity graph,

which is designed to model local interactions primarily based on geographical closeness. Given the coordinates ρ_i and ρ_j for nodes i and j , we calculate the Euclidean distance $d_{ij} = \|\rho_i - \rho_j\|$. The edge weight $\mathbf{A}_p[i, j]$ between these nodes is then computed using a Gaussian kernel, emphasizing nearby connections and diminishing influence with distance:

$$\mathbf{A}_p[i, j] = \exp\left(-\frac{d_{ij}^2}{\tau_d^2}\right), \quad \text{if } d_{ij} \leq \delta_d \text{ else } 0, \quad (15)$$

where τ_d acts as a bandwidth parameter controlling the spatial decay, and δ_d is a threshold to enforce sparsity by limiting connections to within a certain range. This process yields a static, weighted adjacency matrix $\mathbf{A}_p \in \mathbb{R}^{N \times N}$.

2) *Causal-Aware Graph*: In addition to the static graph, we also incorporate a dynamic graph to better capture spatial dependencies that evolve over time. This dynamic graph encodes similarities in the temporal patterns of AQI distributions as well as causal dependencies derived from meteorological dynamics. The final causal-aware graph is computed as a weighted combination of these three components:

$$\begin{aligned} \mathbf{A}_c^t[i, j] = & \frac{1}{W_{total}} (W_t \text{CosSim}(T_i, T_j) \\ & + W_s \text{CosSim}(S_{enh,i}, S_{enh,j}) \\ & + W_c C_{ij}(t)), \end{aligned} \quad (16)$$

where $\text{cosSim}(T_i, T_j)$ denotes the cosine similarity between the entire trend vectors $T_i = [T_i(1), \dots, T_i(L_h)]$ and T_j (similarly for the enhanced seasonal vectors $S_{enh,i}, S_{enh,j}$). $C_{ij}(t)$ is the dynamic causal strength obtained from (14). The terms W_t, W_s, W_c are learnable weights, and $W_{total} = W_t + W_s + W_c$ ensures normalization. Finally, the dynamic causal graph at time step t is denoted as $\mathbf{A}_c^t \in \mathbb{R}^{N \times N}$.

3) *GCN for Dynamic Spatiotemporal Correlation Modeling*: Graph Convolutional Networks (GCNs) have been widely applied to spatiotemporal dynamic modeling. In this work, we employ GCNs to aggregate features across different monitoring stations, capturing spatial correlations in the error signals for improved prediction accuracy. A standard graph convolution layer updates node representations based on their neighbors, typically formulated as $\text{GCN}(H^{(l)}, A) = \sigma(\tilde{A} H^{(l)} W_g^{(l)})$, where $\tilde{A}$ represents the normalized adjacency matrix A , σ represents an activation function such as ReLU, and $W_g^{(l)}$ denotes learnable parameters. GCN is used to aggregate the residual features of nodes $\mathbf{R} \in \mathbb{R}^{T \times N \times c}$ based on the two constructed graphs. The specific process is as follows,

$$\mathbf{X}_{p,t} = \text{GCN}_p(\mathbf{R}_t, \mathbf{A}_p) \quad (17)$$

$$\mathbf{X}_{c,t} = \text{GCN}_c(\mathbf{R}_t, \mathbf{A}_c^t) \quad (18)$$

Here, $\mathbf{R}_t \in \mathbb{R}^{N \times c}$ is the input feature slice at time t . GCN_p leverages the static A_p to capture proximity-based patterns, while GCN_c uses the dynamic A_c at each time step t to incorporate causal influences. Both GCN modules output feature representations $\mathbf{X}_{p,t}, \mathbf{X}_{c,t} \in \mathbb{R}^{N \times d_c}$, where d_c denotes the GCN output feature dimension.

Graph Fusion: Finally, the features extracted from both spatial perspectives are fused at each time step t . This fusion is achieved using a predefined coefficient $\alpha \in [0, 1]$, allowing the model to adaptively weigh the importance of physical proximity versus causal influence case by case:

$$\tilde{\mathbf{X}}_t = \alpha \mathbf{X}_{p,t} + (1 - \alpha) \mathbf{X}_{c,t} \quad (19)$$

For T time steps, the fused spatial feature embedding is denoted as $\tilde{\mathbf{X}}_s \in \mathbb{R}^{L_h \times N \times d_c}$.

D. Patch Mixer for Temporal Dynamics

We propose a Patch Mixer strategy to capture the temporal dynamics of AQI. We combine the spatial feature tensor $\tilde{\mathbf{X}}_s \in \mathbb{R}^{L_h \times N \times c'}$, the trend component $\mathbf{T} \in \mathbb{R}^{L_h \times N \times c}$, and the enhanced seasonal component $\mathbf{S}_{enh} \in \mathbb{R}^{L_h \times N \times c}$. We combine these three tensors and concatenate them along the feature dimension, and the output is denoted as $\mathbf{X}_{in} \in \mathbb{R}^{L_h \times N \times D_{in}}$, where L_h is the input sequence length, N is the number of stations, and D_{in} is the aggregated feature dimension.

1) *Patching*: We first divide the input sequence $\mathbf{X}_{in}$ into P overlapping patches. Each patch has a length of L_p and is extracted with a stride of s_p . The number of generated patches, P , is computed as follows:

$$P = \lfloor (L_h - L_p) / s_p \rfloor + 1 \quad (20)$$

Subsequently, we project each extracted patch into a d -dimensional embedding space. This step transforms the raw patches into dense vector representations, yielding the initial patch embeddings $\mathbf{H}^0 \in \mathbb{R}^{P \times N \times d}$.

2) *Patch-Mixer Module*: The Patch-Mixer Module consists of L_t layers, each composed of an intra-patch layer and an inter-patch layer. These components are responsible for capturing correlations within patches and between patches, respectively. The l -th input is denoted as $\mathbf{H}^l \in \mathbb{R}^{P \times N \times d_l}$ with initial input $\mathbf{H}^0 \in \mathbb{R}^{P \times N \times d}$.

Inter-patch Mixer: In each Mixer module, the input tensor $\mathbf{H}^{(l)}$ is first normalized via batch normalization, denoted as $\text{BN}(\cdot)$. The normalized tensor is then fed into the inter-patch Mixer submodule, where convolution operations are applied along the patch sequence dimension to capture dependencies between patches. The process can be denoted as follows,

$$\mathbf{H}^{l,s} = \text{MLP}(\sigma(\text{BN}(\mathbf{H}^l) \odot \mathbf{W}_d + \mathbf{b}_d)) \quad (21)$$

where $\mathbf{H}^{(l+1)}$ is the output. $\mathbf{W}_d$ and $\mathbf{b}_d$ are learnable parameters. σ denotes an activation function.

Intra-patch Mixer: After inter-patch mixing, the tensor $\mathbf{H}^{l,s}$ is processed by the intra-patch Mixer submodule to capture local feature dependencies. It is first permuted and normalized via batch normalization. The normalized input then passes through an activated linear projection layer, followed by a residual connection that combines the output with the original patch input $\mathbf{H}^l$, producing the final output of the intra-patch block. The overall process is summarized as:

$$\mathbf{H}^{l+1} = \mathbf{H}^{l,s} + \text{MLP}(\sigma(\text{BN}(\text{Permute}(\mathbf{H}^{l,s})))) \quad (22)$$

Algorithm 1: MADGCN for Long-term AQI Prediction.**Require:** Observed AQI $\mathbf{X}$, observed meteorological data $\mathbf{Z}$.**Ensure:** Predicted AQI $\hat{\mathbf{Y}}$.

```

1: 1. Time Series Decomposition
2: EnhancedSTL( $\mathbf{X}$ )  $\rightarrow$  ( $\mathbf{T}, \mathbf{S}_{\text{enh}}, \mathbf{R}$ )
3: 2. Dynamic Causality Discovery
4: Compute  $\mathbf{C}(t)$  from  $\mathbf{R}, \mathbf{Z}$  using DynamicCausality.
5: 3. Causal-aware Graph Convolution
6: Define physical proximity graph  $\mathbf{A}_p$ .
7: for  $t = 1, \dots, T$  do
8:   Define causal-aware graph  $\mathbf{A}_c^t$  using  $\mathbf{T}, \mathbf{S}_{\text{enh}}, \mathbf{C}(t)$ .
9:    $\text{GCN}_p(\mathbf{R}_t, \mathbf{A}_p) \rightarrow \mathbf{X}_{p,t}$ 
10:   $\text{GCN}_c(\mathbf{R}_t, \mathbf{A}_c^t) \rightarrow \mathbf{X}_{c,t}$ 
11:   $\tilde{\mathbf{X}}_t = \alpha \mathbf{X}_{p,t} + (1 - \alpha) \mathbf{X}_{c,t}$ 
12: end for
13:  $\tilde{\mathbf{X}}_s = [\tilde{\mathbf{X}}_1, \dots, \tilde{\mathbf{X}}_T]$ 
14: 4. Patch Mixer for Temporal Dynamics
15: Form  $\mathbf{X}_{in}$  by concatenating  $\tilde{\mathbf{X}}_s, \mathbf{T}$ , and  $\mathbf{S}_{\text{enh}}$ .
16: Generate initial patch embeddings  $\mathbf{H}^{(0)}$  from  $\mathbf{X}_{in}$ .
17: Initialize  $\mathbf{H} = \mathbf{H}^{(0)}$ .
18: for  $l = 1 \dots L_t$  do
19:    $\mathbf{H}_s = \text{MLP}(\sigma(\text{BN}(\mathbf{H}) \odot \mathbf{W}_d + \mathbf{b}_d))$ 
20:    $\mathbf{H} = \mathbf{H}_s + \text{MLP}(\sigma(\text{BN}(\text{Permute}(\mathbf{H}_s))))$ 
21: end for
22: 5. Generate Future Prediction
23:  $\mathbf{H}_o = \text{Add\&Norm}(\mathbf{H})$ 
24:  $\text{MLP}(\text{Flatten}(\mathbf{H}_o)) \rightarrow \hat{\mathbf{Y}}$ 
25: return  $\hat{\mathbf{Y}}$ .
```

Final Prediction: After L_t patch mixer layers, the resulting hidden representation $\mathbf{H}_o$ is fed into a single fully connected layer, which serves as the decoder to predict future values.

$$\hat{\mathbf{Y}} = \text{MLP}(\text{Flatten}(\mathbf{H}_o)) \quad (23)$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{L_f \times N \times c}$ represents the AQI in the next L_f time steps.

V. EXPERIMENT

A. Dataset

1) *Long-range Air Quality Dataset-LargeAQ:* We introduce LargeAQ, a large-scale dataset that records nationwide air quality measurements over multiple years, covering two key types of information: PM2.5 concentrations and meteorological factors.

① **PM2.5 data.** The PM2.5 data used in this study are obtained from the real-time air quality monitoring platform operated by the China National Environmental Monitoring Center (CNEMC), which reports hourly pollutant measurements from a nationwide network of urban and meteorological stations. The dataset covers 343 major Chinese cities and includes 1,341 monitoring stations in total, whose spatial distribution is shown in Fig. 1(c). It spans eight years, from January 1, 2016 to December 31, 2023, at an hourly resolution, resulting in more than

70,000 time steps. PM2.5 concentrations exhibit clear seasonal concept drift. As an illustration, we present a case study in the Yangtze River Delta region and visualize monthly mean PM2.5 levels from September 2022 to August 2023 (Fig. 4). The results show relatively higher pollution in January–April, followed by a noticeable decrease in subsequent months.

② **Meteorological data.** The meteorological data are mainly provided by the China Meteorological Administration and the Nanjing University of Information Science and Technology, supplemented by ERA5 climate reanalysis data from the European Centre for Medium-Range Weather Forecasts (ECMWF) and open data from the U.S. National Climate Data Center (NCDC) to fill in missing values. The final dataset was systematically extracted and refined from a large-scale archive encompassing over 10 years of continuous monitoring data collected from 2,000 real, geographically distributed stations, resulting in a high-quality dataset with complete observations across all measured variables. The LargeAQ dataset integrates crucial meteorological factors: temperature, humidity, precipitation, wind direction, and wind speed. These external factors significantly impact air quality and were processed as follows to harness this information effectively:

- *Temperature and Humidity:* Used directly as fundamental parameters reflecting atmospheric stability and diffusion conditions.
- *Precipitation Treatment:* Processed by computing the cumulative rainfall over a 24-hour period to represent rainfall intensity and its wet deposition effect accurately for long-term prediction.
- *Wind Direction and Speed UV Transformation:* Converted into u (east-west) and v (north-south) components to mitigate angular discontinuity and provide a continuous representation of the wind field for modeling pollutant dispersion.

2) *Another Dataset-KnowAir:* To validate the generalizability of the models, we also incorporate the open-source KnowAir dataset [36]. This dataset contains meteorological and air quality data for 184 Chinese cities spanning 4 years (2015–2018) with a 3-hour temporal resolution. As described by the source, we utilize its three distinct subdatasets for evaluation:

- The **Standard** subdataset contains long-term data across all seasons, split into training, validation, and testing sets in a 2:1:1 ratio to assess general prediction capability.
- The **Heating** subdataset focuses exclusively on the winter heating season, presenting a challenging scenario due to emission changes.
- The **Volatile** subdataset features high-variance pollution patterns simulating an online prediction system, representing the most challenging task.

B. Experiment Setup

To evaluate the model's effectiveness under concept drift, the LargeAQ dataset is split into training, validation, and test sets in a 2:1:1 ratio, following the temporal order of the data. The models are implemented using PyTorch 2.2.0, and GNNs are constructed with the PyTorch Geometric library. Models are run

on an NVIDIA A100 GPU with 40 GB of memory. We adopt the Adam optimizer [46] with an initial learning rate of 0.02 and weight decay of 0.004, coupled with a MultiStepLR learning rate scheduler. Models are trained for a maximum of 300 epochs. Early stopping is implemented with a patience of 20 epochs based on validation loss to prevent overfitting.

The batch size is set to 16 for the *LargeAQ* dataset and 32 for the *KnowAir* dataset. To predict $L_f = 96$ time steps, we use a look-back window L_h of 96 time steps. Our MADGCN consists of $L_t = 2$ patch-mixer layers. The input sequence is segmented into patches of length $L_p = 16$ with a stride $s_p = 8$. PM2.5 concentrations and processed meteorological factors are normalized using Z-score standardization. The mean and standard deviation for normalization are computed solely from the respective training sets to prevent data leakage. Two commonly used metrics, MAE (Mean Absolute Error) and RMSE (Root Mean Squared Error), are used to evaluate model performance.

C. Baseline

To comprehensively evaluate the proposed method for PM2.5 prediction, we compare it with representative baselines spanning three categories: time-series forecasting models, spatiotemporal graph convolutional models, and air-quality-specific forecasting models.

① **Time-series forecasting models.** (1) **PatchTST** [47] applies a Transformer to capture long-range dependencies. (2) **PITS** [48] uses a patch-independent MLP that processes patches separately before aggregation to reduce complexity. (3) **Leddham** [49] employs learnable decomposition to model both intra- and inter-sequence variations. (4) **SparseTSF** [25] leverages cross-period sparsity to decouple periodicity and trend for forecasting; (5) **PDF** [50] is a periodicity decoupling framework for long-term series forecasting.

② **Spatiotemporal graph convolutional models.** (6) **GWNet** combines graph convolutions with dilated causal convolutions to model spatiotemporal dependencies on a fixed graph. (7) **STGCN** [51] integrates GCNs with temporal gated convolutions for spatiotemporal feature extraction. (8) **DCRNN** [52] couples diffusion graph convolution with RNNs to capture spatiotemporal dynamics. (9) **DSTAGNN** [53] models fixed spatial patterns while dynamically reweighting temporal dependencies. (10) **DGCRN** [54] combines dynamic graph construction/convolution with RNNs to capture time-varying relations. (11) **AGCRN** [55] integrates adaptive graph convolution with RNNs to learn node-specific spatial dependencies. (12) **GC-LSTM** [56] embeds graph convolution within an LSTM backbone; (13) **STSGNN** [8] designs specialized 2D graph spectral filters for spatial-temporal graph modeling.

③ **Air-quality-specific forecasting models.** (14) **PM2.5-GNN** [36] augments fixed-graph convolutions with domain knowledge. (15) **MGSFformer** [57] handles irregular sampling intervals and correlations via multi-granularity spatiotemporal fusion blocks. (16) **GAGNN** [58] leverages hierarchical

forecasting and group graphs to capture both local and long-range spatial correlations. (17) **Airformer** [1] preserves fine-grained local details and highlights salient spatiotemporal patterns through enhancement modules. (18) **Air-DualODE** [40] incorporates a discretized, boundary-aware advection–diffusion equation to model pollutant transport.

Specifically, for baseline models that did not originally incorporate meteorological inputs, we concatenate the meteorological factors with the AQI features along the input channel dimension. For models that do incorporate meteorological inputs, we adhere to their original integration mechanism.

D. Overall Performance Comparison on LargeAQ

Table III presents comprehensive experimental results on the LargeAQ dataset across multiple prediction horizons.

① **vs. STGNNs.** STGCN and DCRNN underperform because they rely on static proximity graphs and cannot faithfully capture evolving spatiotemporal dependencies. GWNet and AGCRN improve by learning adaptive graphs, and DSTAGNN further benefits from a temporally dynamic graph. However, these STGNNs remain limited for long-horizon forecasting: they typically model temporal dynamics at a single scale, making them less robust to concept drift, and they incorporate meteorological factors only as auxiliary inputs rather than as part of the dependency learning mechanism.

② **vs. Time-series forecasting models.** SparseTSF is efficient but sacrifices accuracy. PatchTST and Leddam adopt channel-independent modeling with patch-based Transformers, where Leddam further enhances representation learning via dual attention. PITS uses patch-wise MLP modeling with unmasked reconstruction pretraining and complementary contrastive learning to capture local temporal patterns. Overall, pure sequence models often outperform STGNNs on long horizons, but they do not explicitly model spatial/node interactions, which can limit their ability to capture structured dependencies in multivariate urban sensing data.

③ **vs. AQI-specific models.** GAGNN targets national-scale forecasting with hierarchical spatial modeling, but learning reliable hierarchies over large urban networks is challenging and can lead to suboptimal results. PM2.5-GNN couples GNNs with GRUs and uses meteorological factors as auxiliary features, yielding stronger AQI dynamics modeling. AirFormer further advances performance with a two-stage design that combines deterministic attention-based encoding with stochastic latent factors for uncertainty modeling.

Summary: Our method consistently achieves the best results across all horizons. For 24-hour prediction, **MADGCN** attains **18.57/27.79** MAE/RMSE, reducing MAE by 5.3% over PITS and 8.3% over PM2.5-GNN. It also degrades much more gracefully with horizon length, with only a **41.2%** MAE increase from 12h to 96h. These gains stem from (i) a **dynamic dual-graph** that jointly captures physical proximity and meteorological causality via **Granger causality**, and (ii) **patch-based temporal evolution learning** that models long-range dynamics by processing fused spatiotemporal features across both temporal patches and spatial channels.

TABLE III
LONG-TERM PREDICTION PERFORMANCE ON THE LARGE AQ DATASET. THE BEST RESULTS ARE **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED IN **BLUE**.

Model	12h		24h		36h		48h		60h		72h		96h	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
<i>Time Series Prediction Model</i>														
SparseTSF	18.24	27.12	20.75	31.18	21.88	32.22	22.43	33.31	23.01	33.58	23.76	34.15	24.15	34.83
Leddarn	17.25	25.87	20.15	30.05	21.05	31.15	21.75	32.15	22.28	32.42	22.73	32.98	23.22	33.58
PatchTST	16.72	24.73	<u>19.55</u>	<u>28.88</u>	20.78	30.53	21.67	31.67	22.32	32.43	22.81	33.05	23.25	33.60
PITS	<u>16.69</u>	<u>24.57</u>	19.62	28.97	<u>20.73</u>	<u>30.41</u>	<u>21.42</u>	<u>31.44</u>	<u>22.25</u>	<u>32.35</u>	<u>22.50</u>	<u>32.97</u>	<u>23.18</u>	<u>33.52</u>
PDF	16.88	24.71	19.63	28.92	20.75	30.48	21.35	31.42	22.13	32.21	22.42	32.95	23.05	33.38
<i>Spatiotemporal Graph Convolutional Model</i>														
STGCN	21.83	32.87	25.55	38.32	27.37	40.73	28.67	42.47	29.47	43.63	30.12	44.57	30.67	45.41
DCRNN	20.32	30.37	23.75	35.44	25.43	37.67	26.63	39.23	27.42	40.32	28.01	41.17	28.50	41.92
GWNet	18.28	27.07	21.33	31.56	22.83	33.52	23.93	34.97	24.67	36.13	25.21	37.15	25.69	38.11
GC-LSTM	20.51	30.83	23.99	36.01	25.67	38.27	26.87	39.87	27.63	41.03	28.23	41.92	28.79	42.70
AGCRN	20.46	30.71	23.87	35.95	25.55	38.12	26.79	39.74	27.51	40.91	28.16	41.85	28.67	42.57
DGCRN	19.73	29.43	23.07	34.31	24.67	36.47	25.83	38.03	26.57	39.17	27.15	40.12	27.69	40.98
DSTAGNN	18.17	27.48	21.19	32.02	22.63	33.87	23.67	35.17	24.37	36.08	24.92	36.79	25.43	37.46
STSGNN	20.35	30.55	23.65	35.72	25.62	38.25	26.65	39.55	27.60	41.05	28.05	41.68	28.75	42.66
<i>Specific Air Pollution Forecasting</i>														
Airformer	17.08	25.43	19.93	29.66	21.28	31.37	22.27	32.53	23.03	33.32	23.61	33.93	24.15	34.49
PM2.5-GNN	17.36	26.03	20.25	30.32	21.63	32.13	22.65	33.37	23.32	34.28	23.83	35.06	24.30	35.77
Air-DualODE	18.21	27.45	21.26	31.74	22.78	34.01	23.67	35.31	24.59	36.02	25.13	37.01	25.81	38.01
MGSFformer	17.68	26.35	20.58	30.52	21.52	31.68	22.23	32.68	22.76	32.95	23.21	33.52	23.71	34.12
GAGNN	21.25	32.05	24.82	37.28	26.79	39.65	27.45	41.53	28.72	42.79	29.46	43.32	30.02	44.28
MADGCN (Ours)	15.83	23.67	18.57	27.79	19.76	29.42	20.63	30.53	21.28	31.37	21.85	32.18	22.36	32.94

E. Ablation Study

In this section, we evaluate the effectiveness of each component by constructing ablation variants as follows:

- **w/o Graph** means that we remove the graph convolutional network.
- **w/o PPG** means that we remove the physical proximity graph.
- **w/o CAG** means that we remove three graphs from the causal-aware graph convolutional network.
- **w/o Meteorology** indicates that we remove the Causal-Aware Graph.
- **w/o STL** means that we remove the seasonal-trend decomposition module.
- **w/o Cycle** means that we remove the seasonal component enhancement module with the CyclConv network.

As shown in Fig. 5, we observe that all ablated variants perform worse than the full model to varying degrees, confirming the effectiveness of each component. The variant without graph modeling (w/o Graph) achieves the worst performance, highlighting the necessity of incorporating spatial dependencies. This is because inter-node relationships provide valuable contextual information that aids accurate prediction. Both w/o

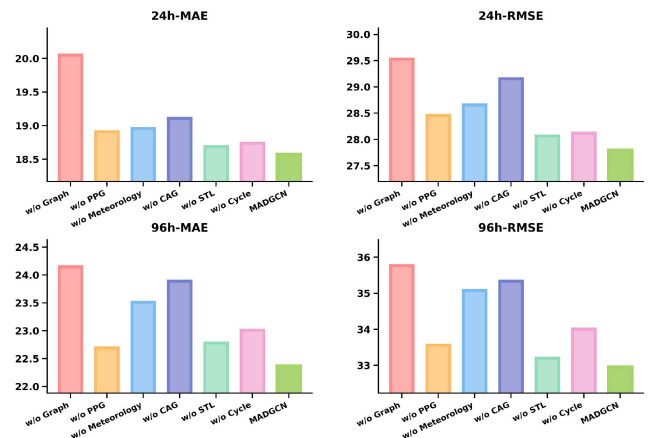


Fig. 5. Ablation results on the LargeAQ dataset.

PPG and w/o CAG result in significantly higher prediction errors, demonstrating the importance of modeling spatiotemporal dependencies from multiple perspectives. In particular, the w/o Meteorology variant—which excludes dynamic meteorological factors—shows the second-worst performance, indicating that

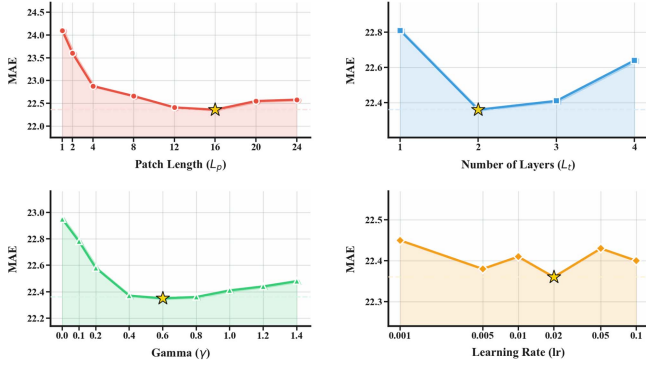


Fig. 6. Key hyperparameter sensitivity analysis.

TABLE IV
MODEL EFFICIENCY COMPARISON

Model	Memory (MB)	Train Time (s/epoch)	Infer Time (s/tier)	MAE (AVG)
GWNet	24672 (3.77 $\times$)	86.45 (10.74 $\times$)	15.75 (10.71 $\times$)	23.13 (+15.4%)
AGCRN	26564 (4.06 $\times$)	113.4 (14.09 $\times$)	23.80 (16.19 $\times$)	25.86 (+29.0%)
PM2.5-GNN	34094 (5.21 $\times$)	31.5 (3.91 $\times$)	7.98 (5.43 $\times$)	21.91 (+9.3%)
AirFormer	30362 (4.64 $\times$)	68.6 (8.52 $\times$)	12.81 (8.71 $\times$)	21.62 (+7.9%)
Ours	6540	8.05	1.47	20.04

integrating meteorological data is essential for capturing meaningful spatiotemporal dynamics in AQI prediction. The poor performance of w/o STL further supports our design: temporal decomposition allows the model to separately learn from trend, seasonal, and residual components, leading to more precise and structured temporal modeling.

F. Hyperparameter Sensitivity Analysis

We systematically evaluate the impact of key hyperparameters in Fig. 6, including patch length (L_p), number of layers (L_t), STL enhancement factor (γ), and learning rate. The results show that $L_p = 16$ achieves the best performance, as it effectively captures temporal context without introducing redundancy. A low value of γ fails to sufficiently enhance the seasonal component, thereby limiting the model's ability to capture periodic patterns. In contrast, excessive enhancement introduces noise and distorts the original seasonal signal, leading to degraded forecasting accuracy. The optimal setting for L_t is found to be 2, which strikes a good balance between model capacity and the risk of overfitting. When L_t increases beyond this point, the model becomes more prone to overfitting, resulting in performance deterioration.

G. Efficiency Analysis

We further evaluated the efficiency of the proposed model, and the results are presented in Table IV. Compared with advanced air quality forecasting models, MADGCN has lower parameter complexity and higher time efficiency. In particular, compared with the competitive AirFormer model, our parameter complexity is reduced by 3.64 $\times$, while the training time and inference time are reduced by 7.52 $\times$ and 7.71 $\times$, respectively, all while maintaining high prediction accuracy.

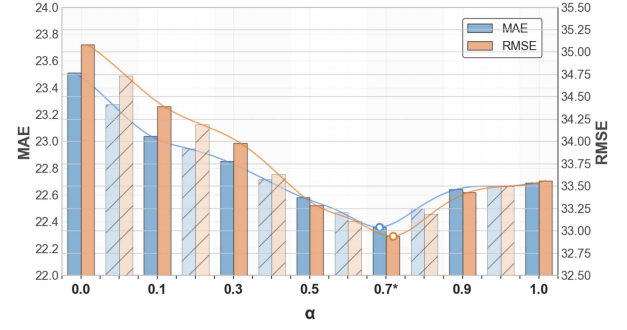
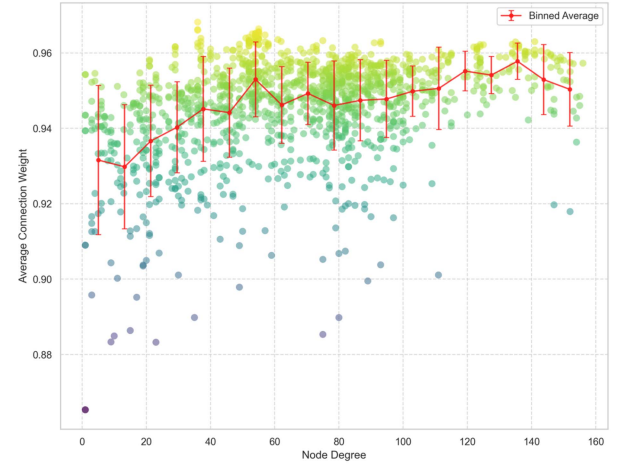
Fig. 7. Impact of fusion weight α on LargeAQ.

Fig. 8. Average connection weight and node degree distribution.

H. Causal Graph Convolution

We also investigate the impact of the graph fusion weight α , which balances the contributions of the two graph components: the Physical Proximity Graph (PPG), weighted by $1 - \alpha$, and the Causal-Aware Graph (CAG), weighted by α . As shown in Fig. 7, the 96-hour prediction results on LargeAQ demonstrate that the model performance is sensitive to the balance between PPG and CAG features. The results exhibit a clear unimodal trend with respect to α , where optimal performance is achieved when α is approximately 0.7—indicating that assigning 70% weight to the Causal-Aware Graph and 30% to the Physical Proximity Graph yields the best trade-off for long-term prediction. Further analysis reveals that the optimal value of α increases as the prediction horizon extends, suggesting a growing reliance on causal relationships inferred from meteorological factors and periodic pattern similarities in longer-term predictions.

1) *Average Degree Distribution of Nodes in the Causal Graph:* Fig. 8 visualizes the learned connection weights under the optimal setting. We observe a strong positive correlation between node degree and average connection weight—nodes with higher degrees (i.e., hub nodes) tend to form stronger connections on average. This structural characteristic plays a critical role in improving prediction accuracy, especially for

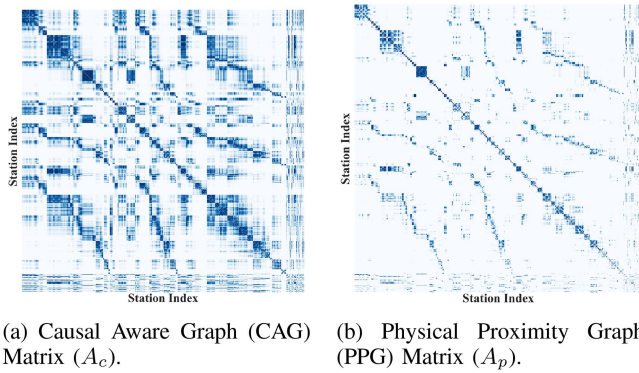


Fig. 9. Visual comparison of adjacency matrices.

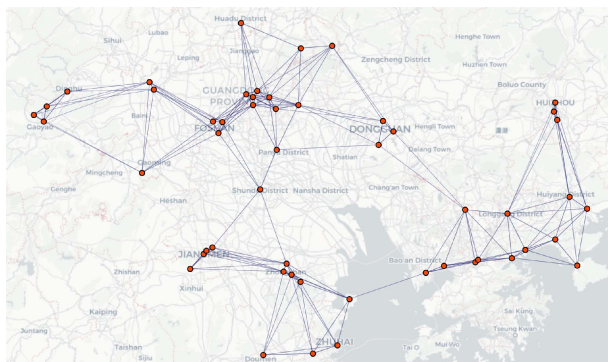


Fig. 10. Edge weight visualization of the causal graph in the Pearl River Delta.

long-horizon prediction. By assigning higher weights to edges involving high-degree and influential nodes, our model prioritizes the propagation of important, often causally driven signals across the network. This enables more effective modeling of large-scale spatiotemporal dependencies that are essential for accurate air pollution forecasting, thereby enhancing both the reliability and precision of the model's forecasts.

2) *Graph Edge Weight Visualization*: We visualize two graphs in Figs. 9 and 10. The Physical Proximity Graph reflects distance-based connectivity and exhibits a block structure corresponding to geographically clustered stations. This locality is consistent with its objective of modeling geographic proximity. In contrast, the Causal-Aware Graph shows a more diffuse connectivity pattern shaped by meteorological factors, such as wind fields that govern pollutant transport and temperature that modulates chemical reaction rates. A prior study [59] reported that causal interactions are closely tied to spatial proximity. In our setting, however, meteorological factors induce long-range, non-local associations. Consequently, the Causal-Aware Graph identifies dependencies that extend beyond geometric closeness. These distant edges, which distance-based Physical Proximity Graphs typically miss, are often driven by large-scale meteorological systems or synchronized responses to regional atmospheric conditions. This indicates that the Causal-Aware

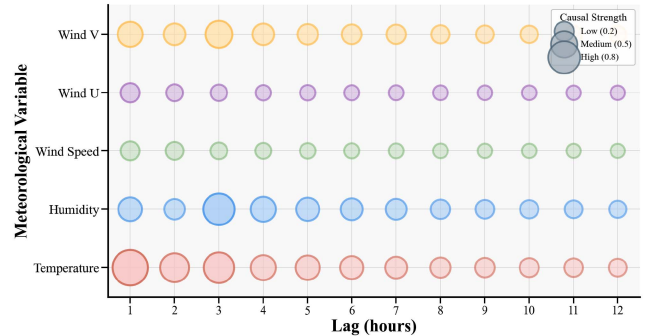


Fig. 11. Granger causality analysis of meteorological factors.

Graph captures causal relationships that are not limited to spatial adjacency.

Fig. 10 further illustrates the edge visualization of the Causal-Aware Graph, where edges with weights greater than 0.9 in the graph are mapped onto the Pearl River Delta region. The visualization highlights two key observations: Many strong causal connections exhibit geographic consistency, occurring within densely populated areas or between neighboring cities. This is likely due to similar meteorological conditions (e.g., temperature) across these regions. The graph also reveals numerous non-local and long-range connections, such as those between Zhongshan and Zhuhai. These links may result from complex environmental factors, such as wind direction, that facilitate pollutant transport over large distances. Taken together, this visual analysis demonstrates the effectiveness of our dual-graph strategy in capturing the intricate spatiotemporal dependencies present in air quality data.

3) *Meteorological Variable Importance Analysis*: To provide deeper insight into the “meteorology-aware” aspect of our model, we analyze which meteorological factors are most influential in forming the causal graph edges. Fig. 11 presents a Granger causality analysis of meteorological variables on PM_{2.5}.

The analysis reveals a clear hierarchy of meteorological influence: Temperature dominates short-term causal effects, particularly at 1–5-hour lags, which aligns with the physical mechanism where temperature inversions trap pollutants near the surface. Humidity influences secondary aerosol formation at 2–4-hour lags, and Wind V (north-south component) reflects synoptic-scale pollution transport patterns. These findings directly inform the edge formation in our dynamic causal graph, enabling variable-specific, lag-aware spatial dependency modeling.

1. Performance Comparison for High-Pollution Patterns

In this section, we conduct an in-depth analysis of model performance under high-pollution regimes, which are typically of greater practical concern. We evaluate our method on the KnowAir dataset as well as three time-specific subsets.

The results are summarized in Table V. It is evident that most existing models perform poorly under high-pollution

TABLE V
AVERAGE PREDICTION PERFORMANCE ACROSS TIME PERIODS ON THE KNOWAIR DATASET, WITH PARTICULAR EMPHASIS ON HIGH-POLLUTION PATTERNS. Δ DENOTES THE PERCENTAGE DECREASE IN PERFORMANCE RELATIVE TO OUR MODEL.

Dataset	Models	1–24 h		24–48 h		48–72 h		Average over 72 h			
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	Δ	RMSE	Δ
Standard	PITS	<u>10.41</u>	<u>16.27</u>	12.59	19.73	13.73	21.08	12.24	+13.6%	19.03	+11.8%
	PatchTST	10.46	16.41	12.64	19.78	13.98	21.23	12.36	+14.8%	19.14	+12.5%
	DCRNN	12.01	18.80	14.03	21.65	13.58	22.32	13.21	+22.6%	20.92	+22.9%
	GWNet	11.95	18.92	14.18	21.78	14.05	22.45	13.39	+24.3%	21.05	+23.7%
	DSTAGNN	11.53	18.25	13.98	21.00	14.19	21.18	13.23	+22.8%	20.14	+18.3%
	PM2.5-GNN	10.88	17.40	12.89	20.09	13.50	20.82	12.42	+15.3%	19.44	+14.2%
	Airformer	10.82	17.27	12.79	20.13	<u>13.11</u>	<u>19.95</u>	12.24	+13.6%	19.12	+12.3%
	MGSFformer	10.62	16.67	<u>12.29</u>	<u>19.28</u>	13.24	20.22	<u>12.05</u>	+11.9%	<u>18.72</u>	+10.0%
	Ours	9.53	15.07	11.12	17.57	11.66	18.42	10.77	–	17.02	–
Heating	PITS	<u>16.09</u>	<u>23.34</u>	20.41	28.94	22.05	31.82	19.52	+8.0%	28.03	+9.9%
	PatchTST	16.18	23.39	20.46	28.99	21.70	31.37	19.45	+7.6%	27.92	+9.4%
	DCRNN	19.41	27.83	22.28	31.46	22.90	32.43	21.53	+19.1%	30.57	+19.8%
	GWNet	18.75	27.45	22.85	32.18	23.72	33.65	21.77	+20.4%	31.09	+21.9%
	DSTAGNN	17.80	25.58	22.36	31.85	23.46	33.35	21.21	+17.3%	30.26	+18.6%
	PM2.5-GNN	16.89	24.84	20.91	29.71	21.66	30.98	19.82	+9.6%	28.51	+11.8%
	Airformer	16.54	24.64	20.61	29.34	<u>21.35</u>	<u>30.24</u>	19.50	+7.9%	28.07	+10.0%
	MGSFformer	16.34	24.24	<u>20.11</u>	<u>28.49</u>	21.55	30.47	<u>19.33</u>	+6.9%	<u>27.73</u>	+8.7%
	Ours	15.25	22.05	18.93	26.80	20.07	27.68	18.08	–	25.51	–
Volatile	PITS	<u>21.66</u>	<u>31.96</u>	26.77	38.02	29.28	40.75	25.90	+7.5%	36.91	+7.4%
	PatchTST	21.71	32.31	26.52	38.07	29.33	40.80	25.85	+7.3%	37.06	+7.8%
	DCRNN	25.99	36.18	29.96	42.43	31.71	43.67	29.22	+21.2%	40.76	+18.6%
	GWNet	26.18	37.25	31.42	44.76	33.85	47.12	30.48	+26.5%	43.04	+25.2%
	DSTAGNN	23.47	35.37	28.68	40.07	29.92	40.27	27.36	+13.5%	38.57	+12.2%
	PM2.5-GNN	22.34	33.28	26.93	38.28	28.54	40.35	25.94	+7.6%	37.30	+8.5%
	Airformer	22.06	32.96	26.97	38.51	28.23	39.10	25.75	+6.8%	36.86	+7.2%
	MGSFformer	21.86	32.66	<u>26.47</u>	<u>37.66</u>	<u>28.03</u>	<u>38.94</u>	<u>25.45</u>	+5.6%	<u>36.42</u>	+6.0%
	Ours	20.17	30.57	25.29	35.96	26.85	36.59	24.10	–	34.37	–

scenarios, indicating their limitations in capturing extreme spatiotemporal dynamics. In particular, the heating subset exhibits more complex spatiotemporal dependencies due to seasonal emission sources and meteorology-driven dispersion patterns. Compared with the best-performing baseline, MGSFformer (MAE/RMSE: 19.33/27.73), our model achieves MAE/RMSE scores of 18.08/25.51, corresponding to improvements of 6.5% and 8.0%, respectively. This demonstrates that the proposed causality-aware dynamic graph discovery mechanism effectively captures the influence of meteorological factors on heating-induced pollution patterns. On the volatile subset, where pollution peaks are highly unpredictable, our model maintains strong robustness, achieving MAE/RMSE scores of 24.10/34.37, outperforming MGSFformer (25.45/36.42) by 5.3% and 5.6%, respectively. In contrast to competing methods whose performance degrades substantially, our approach consistently delivers superior accuracy across all subsets.

J. Cross-Domain Validation on the Energy Dataset

To assess the scalability of our model, we additionally evaluate our method on an energy-domain benchmark, SDWPF [60], which contains power generation along with wind speed and other weather variables. SDWPF spans January 2020 to December 2021 and provides 10-minute SCADA measurements from 134 wind turbines in a single wind farm. We use an

TABLE VI
PERFORMANCE COMPARISON ON THE ENERGY DATASET

Model	6h		12h		24h	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
STGCN	38.45	48.56	39.82	50.13	40.67	51.89
DCRNN	37.12	46.78	38.35	48.21	39.28	49.86
GWNet	35.67	44.93	36.54	46.12	37.23	47.35
GC-LSTM	37.45	47.19	38.56	48.67	39.42	50.15
AGCRN	36.82	46.35	36.45	45.89	36.18	45.52
DGCRN	35.93	45.28	35.62	44.86	35.38	44.53
DSTAGNN	35.42	44.65	35.18	44.32	34.95	44.08
MADGCN	33.86	42.45	33.58	42.12	33.35	41.86

input length of 144 steps (i.e., 24 hours) and forecast the next 6, 12, and 24 hours. We compare MADGCN with representative spatiotemporal graph learning baselines that are known to perform well in short-horizon forecasting. Results in Table VI show that our model achieves consistently strong performance across all horizons, indicating favorable scalability and robustness.

VI. DISCUSSION

We discuss potential directions for future research:

① **LLM.** Recent advances in large language models (LLMs) have demonstrated strong generalization capabilities in time series forecasting tasks [61], [62]. Their ability to encode global patterns and perform zero-shot inference makes them promising

candidates for enhancing environmental prediction systems. As a future direction, we plan to integrate MADGCN with LLMs to improve both accuracy and generalization across unseen regions and pollution patterns.

② Expanding Air Quality Monitoring Graph. Air quality monitoring networks are often dynamic in practice, with new stations being added over time. This introduces challenges for continual learning and model updating without retraining from scratch [63], [64]. In future work, we will investigate online and incremental learning strategies tailored for evolving graph structures, aiming to build a dynamic air monitoring system capable of adapting to network expansions in real time.

VII. CONCLUSION

To enable accurate long-horizon air pollution forecasting, we decouple temporal structures and incorporate meteorological variables as causal covariates. We propose MADGCN, a Meteorology-Aware Decoupled Graph Convolutional Network, which first decomposes the AQI series into trend, seasonal, and residual components. We then introduce a dynamic causality discovery mechanism to capture time-varying causal dependencies between meteorological conditions and AQI evolution. Guided by the inferred causal structure, a causal convolutional network together with a PatchMixer module models pollution propagation by learning spatial interactions and multiscale temporal dynamics. We evaluate MADGCN on two real-world datasets and compare it with 15 state-of-the-art baselines. The results show that MADGCN consistently improves forecasting accuracy over existing methods.

VII. CONTRIBUTION STATEMENT

Binwu Wang is the project leader, responsible for the overall design of the paper, manuscript writing, and partial funding support. Zhiqing Cui contributed to method design, code implementation, and experimental evaluation. Guangjun Wang was responsible for literature review and data collection. Zhengyang Zhou assisted in paper revision and contributed to the writing of the manuscript. Fan Meng and Jingjia Luo contributed to manuscript revision and provided academic supervision throughout the research process. Yang Wang, as the principal investigator of the research group, supervised the overall planning of the project, provided experimental resources and technical support, and contributed partial funding.

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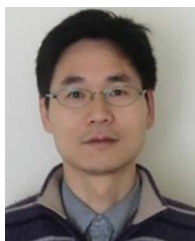


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